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Appendix A: Detailed analysis of the DC-JL model

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NOTATION

Symbol	Name	Description / Mathematical Expression
a	Control initial stiffness	$a = 1/E_0$
b	Control max deviatoric stress	$b = 1/TSS$
E_0	Initial average elastic modulus	It is the slope value of the stress-strain curve in the linear-elastic stage.
l	Second-order softening shape parameter	$l = 1/(E_0 \varepsilon_f^2)$ Controls the rate of decline of the curve during the later softening stage
m	First-order softening shape parameter	$m = 1/TSS - 2/(E_0 \cdot \varepsilon_f)$ Controls the descent rate of the curve during the initial softening stage. The larger the value, the greater the initial descent
p	The plasticity factor	$p = \begin{cases} 1 - (\varepsilon/\varepsilon_s)^2, & \varepsilon \leq \varepsilon_s \\ 0, & \varepsilon > \varepsilon_s \end{cases}$
$P_l(\varepsilon)$	Plasticity term	$P_l(\varepsilon) = \frac{\varepsilon^p}{(1/E_0) \cdot p + (1/TSS) \cdot \varepsilon^p}$
RSS	Residual shear strength	RSS is the residual shear strength at the end of the test
$S_o(\varepsilon)$	Softening term	$S_o(\varepsilon) = \frac{\varepsilon}{(1/E_0) + m\varepsilon + l\varepsilon^2}$
TSS, ε_f	Strength and failure strain	TSS represents the peak or maximum value of the stress-strain curve, and ε_f corresponds to the strain at the TSS point
URS	Ultimate residual shear strength	URS is the residual shear strength when $\varepsilon \rightarrow \infty$
w_i	Comprehensive weight	$0\% \leq w_i \leq 100\%$, which is a comprehensive weighting rate for the plastic and softening terms (see section 2, Figure A.1)
σ_1, σ_3	Major/Minor stress	Vertical stress and confining pressure in the triaxial test
ε	Strain	Axial deformation
ε_s	Yield Strain	The strain point is the transition from linear-elastic deformation to plastic deformation.

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1 Model analysis

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1.1 Duncan-Chang- Jiang-Lu (DC-JL) Model

$$(\sigma_1 - \sigma_3)^* = (1 - w_i) \cdot P_l(\varepsilon) + w_i \cdot S_o(\varepsilon) \quad (A.1)$$

$$\begin{cases} (\sigma_1 - \sigma_3)^* = \frac{1 - w_i}{URS/TSS} \cdot \frac{\varepsilon^p}{a \cdot p + b \cdot \varepsilon^p} + \frac{w_i}{(1 - URS/TSS)} \cdot \frac{\varepsilon}{a + m\varepsilon + l\varepsilon^2} \\ p = \max(1 - (\varepsilon/\varepsilon_s)^2, 0) \end{cases} \quad (A.2)$$

5 where $(\sigma_1 - \sigma_3)^*$ is the prediction value of deviatoric stress, and other symbols are seen in table [NOTATION](#).

6 All Formulations presented in this paper have been carefully verified through repeated manual checking

7 and symbolic computation using the MATLAB symbolic toolbox.

8 1.2 The plasticity item $P_l(\varepsilon)$ analysis

9 The $(1-w_i) \cdot P_l(\varepsilon)$: based on a modified Duncan-Chang's formulation for sand soil:

$$\begin{aligned} (1-w_i) \cdot P_l(\varepsilon) &= (1-w_i) \cdot \frac{\varepsilon^p}{a \cdot p + b \cdot \varepsilon^p} \\ p &= \max(1 - (\varepsilon/\varepsilon_s)^2, 0) \end{aligned} \quad (\text{A.3})$$

10 (1.2.1) Considering the limit of $(1-w_i) \cdot P_l(\varepsilon)$ as $\varepsilon \rightarrow \varepsilon_f$, $\varepsilon_f \geq \varepsilon_s$,

$$\begin{aligned} \text{When } \varepsilon \rightarrow \varepsilon_f, \lim_{\varepsilon \rightarrow \varepsilon_f} p &= 0 \\ \lim_{\varepsilon \rightarrow \varepsilon_f} (1-w_i) \frac{\varepsilon^0}{a \cdot 0 + b \varepsilon^0} &= (1-w_i) \frac{1}{b} \\ &= (1-w_i) \cdot TSS \end{aligned} \quad (\text{A.4})$$

11 (1.2.2) Considering the limit of $(1-w_i) \cdot P_l(\varepsilon)$ as $\varepsilon \rightarrow \infty$

$$\begin{aligned} \text{When } \varepsilon \rightarrow \infty, \lim_{\varepsilon \rightarrow \infty} p &\equiv 0 \\ \lim_{\varepsilon \rightarrow \infty} (1-w_i) \frac{\varepsilon^0}{a \cdot 0 + b \varepsilon^0} &= (1-w_i) \frac{1}{b} \\ &= (1-w_i) \cdot TSS \\ &= URS \end{aligned} \quad (\text{A.5})$$

12 When $w_i=0$ and $p=1$, [Equation \(A.3\)](#) turns to Duncan-Chang's model [[1](#)]

$$\begin{aligned} \lim_{\varepsilon \rightarrow \infty} \frac{\varepsilon}{a + b\varepsilon} &= \frac{1}{b} \\ &= TSS \end{aligned} \quad (\text{A.6})$$

13 (1.2.3) First Derivative function $P'_l(\varepsilon)$: differentiation using the quotient rule:

$$\begin{cases} f(\varepsilon) = \varepsilon^p \\ g(\varepsilon) = a \cdot p + b \cdot \varepsilon^p \\ P'_l(\varepsilon) = [f'(\varepsilon) \cdot g(\varepsilon) - f(\varepsilon) \cdot g'(\varepsilon)] / ([g(\varepsilon)]^2) \end{cases} \quad (\text{A.7})$$

14 Derivative function $f'(\varepsilon)$ and $g'(\varepsilon)$

$$\begin{cases} f'(\varepsilon) = -\frac{2\varepsilon^2 \ln(\varepsilon) - \varepsilon_s^2 + \varepsilon^2}{\varepsilon_s^2 \varepsilon^{(\varepsilon/\varepsilon_s)^2}} \\ g'(\varepsilon) = -\frac{b\varepsilon^2 - b\varepsilon_s^2 + 2a\varepsilon \cdot \varepsilon^{(\varepsilon/\varepsilon_s)^2} + 2b\varepsilon_s^2 \cdot \ln(\varepsilon)}{\varepsilon_s^2 \varepsilon^{(\varepsilon/\varepsilon_s)^2}} \end{cases} \quad (\text{A.8})$$

15 Combining Equations (A.6) and (A.7), we get $P_l'(\varepsilon)$

$$P_l'(\varepsilon) = \frac{a\varepsilon^{(\varepsilon/\varepsilon_s)^2} \cdot [2\varepsilon^4 \ln(\varepsilon) + \varepsilon_s^4 + \varepsilon^4 - 2\varepsilon_s^2 \varepsilon^2 \ln(\varepsilon)]}{(a\varepsilon_s^2 \cdot \varepsilon^{(\varepsilon/\varepsilon_s)^2} - a\varepsilon^2 \varepsilon^{(\varepsilon/\varepsilon_s)^2} + b\varepsilon_s^2 \varepsilon)^2} \quad (\text{A.9})$$

16 (1.2.4) Considering the limit of $(1-w_i) \cdot P_l'(\varepsilon)$ as $\varepsilon \rightarrow 0^+$

$$\begin{aligned} \text{When } \varepsilon \rightarrow 0^+, \lim_{\varepsilon \rightarrow 0^+} \varepsilon^2 \ln(\varepsilon) &= \lim_{\varepsilon \rightarrow 0^+} \frac{\ln(\varepsilon)}{\frac{1}{\varepsilon^2}} \xrightarrow{L'H\acute{o}pital's Rule} \lim_{\varepsilon \rightarrow 0^+} -\varepsilon^2 = 0 \\ \text{When } \varepsilon \rightarrow 0^+, 0^0 &= 1, \lim_{\varepsilon \rightarrow 0^+} \varepsilon^{(\varepsilon/\varepsilon_s)^2} = 1 \\ (1-w_i) \cdot P_l'(\varepsilon) &= (1-w_i) \cdot \frac{a \cdot 1 \cdot [0 + \varepsilon_s^4 + 0 - 0]}{(a\varepsilon_s^2 \cdot 1 - 0 + 0)^2} \\ &= (1-w_i) \cdot \frac{1}{a} \\ &= (1-w_i) \cdot E_0 \end{aligned} \quad (\text{A.10})$$

17 (1.2.5) Extending the limit of $(1-w_i) \cdot P_l'(\varepsilon)$ as $\varepsilon \rightarrow \infty$

$$\begin{aligned} \text{When } \varepsilon \rightarrow \infty, p = 0, \lim_{\varepsilon \rightarrow \infty} P_l(\varepsilon) &\equiv \frac{1}{b} \\ (1-w_i) \cdot \lim_{\varepsilon \rightarrow \infty} P_l'(\varepsilon) &\approx (1-w_i) \cdot \frac{a\varepsilon^{(\varepsilon/\varepsilon_s)^2} \cdot [2\varepsilon^4 \ln(\varepsilon)]}{(-a\varepsilon^2 \varepsilon^{(\varepsilon/\varepsilon_s)^2})^2} \\ &= (1-w_i) \cdot \frac{[2\ln(\varepsilon)]}{a\varepsilon^{(\varepsilon/\varepsilon_s)^2}} \\ &= 0 \end{aligned} \quad (\text{A.11})$$

18 1.3 The softening item $S_o(\varepsilon)$ analysis

19 The $w_i S_o(\varepsilon)$: based on a modified Jiang's formulation for rock.

$$w_i \cdot S_o(\varepsilon) = w_i \cdot \frac{\varepsilon}{a + m\varepsilon + l\varepsilon^2} \quad (\text{A.12})$$

20 When $w_i=1$, Equation (A.12) turns to Jiang's model [2].

21 (1.3.1) Considering the limit of $w_i S_o(\varepsilon)$ as $\varepsilon \rightarrow \varepsilon_f$:

$$\begin{aligned}\lim_{\varepsilon \rightarrow \varepsilon_f} w_i \frac{\varepsilon}{a + m\varepsilon + l\varepsilon^2} &= \lim_{\varepsilon \rightarrow \varepsilon_s} w_i \frac{\frac{\varepsilon_f}{\frac{1}{E_0} + \left(\frac{1}{TSS} - \frac{2}{E_0\varepsilon_f}\right)\varepsilon_f + \frac{1}{E_0\varepsilon_f^2} \cdot \varepsilon_f^2}}{\varepsilon_f} \\ &= w_i \cdot TSS\end{aligned}\quad (\text{A.13})$$

22 (1.3.2) Considering the limit of $w_i S_o(\varepsilon)$ as $\varepsilon \rightarrow \infty$

$$\lim_{\varepsilon \rightarrow \infty} w_i \frac{\varepsilon}{a + m\varepsilon + l\varepsilon^2} \xrightarrow{L'H\acute{o}pital's Rule} \lim_{\varepsilon \rightarrow \infty} w_i \cdot \frac{1}{m + 2l\varepsilon} = 0 \quad (\text{A.14})$$

23 (1.3.3) First Derivative function $S_o'(\varepsilon)$: differentiation using the quotient rule:

$$\begin{aligned}(u/v)' &= (u'v - uv')/v^2, \quad u = \varepsilon, v = a + m\varepsilon + l\varepsilon^2: \\ S_o'(\varepsilon) &= \frac{1 \cdot (a + m\varepsilon + l\varepsilon^2) - \varepsilon(m + 2l\varepsilon)}{(a + m\varepsilon + l\varepsilon^2)^2} \\ &= \frac{(a - l\varepsilon^2)}{(a + m\varepsilon + l\varepsilon^2)^2}\end{aligned}\quad (\text{A.15})$$

24 (1.3.4) Extending the limit of $w_i S_o'(\varepsilon)$ as $\varepsilon \rightarrow 0^+$

$$\lim_{\varepsilon \rightarrow 0^+} w_i \cdot \frac{a - l\varepsilon^2}{(a + m\varepsilon + l\varepsilon^2)^2} = w_i \cdot E_0 \quad (\text{A.16})$$

25 (1.3.5) Considering the limit of $w_i S_o'(\varepsilon)$ as $\varepsilon \rightarrow \infty$

$$\lim_{\varepsilon \rightarrow \infty} w_i \cdot \frac{a - l\varepsilon^2}{(a + m\varepsilon + l\varepsilon^2)^2} = 0 \quad (\text{A.17})$$

26 1.4 DC-JL model analysis

27 (1.4.1) Considering the limit of the DC-JL model as $\varepsilon \rightarrow \varepsilon_f$

28 Combining Equations (A.4) and (A.13), we get:

$$\begin{aligned}\lim_{\varepsilon \rightarrow \varepsilon_f} [(1 - w_i)P_l(\varepsilon) + w_i \cdot S_o(\varepsilon)] &= (1 - w_i) \cdot TSS + w_i \cdot TSS \\ &= TSS\end{aligned}\quad (\text{A.18})$$

29 (1.4.2) Extending the limit of the DC-JL model as $\varepsilon \rightarrow \infty$

30 Combining Equations (A.5) and (A.14), we get:

$$\begin{aligned}\lim_{\varepsilon \rightarrow \infty} [(1 - w_i)P_l(\varepsilon) + w_i \cdot S_o(\varepsilon)] &= (1 - w_i) \cdot TSS + 0 \\ &= (1 - w_i) \cdot TSS \\ &= URS\end{aligned}\quad (\text{A.19})$$

31 **(1.4.3) The first Derivative function of the DC-JL model:**

32 Combining Equations (A.9) and (A.15), we get

$$(1 - w_i) \cdot P_l'(\varepsilon) + w_i \cdot S_o'(\varepsilon) =$$

$$(1 - w_i) \cdot \frac{a\varepsilon^{(\varepsilon/\varepsilon_s)^2} \cdot [2\varepsilon^4 \ln(\varepsilon) + \varepsilon_s^4 + \varepsilon^4 - 2\varepsilon_s^2 \varepsilon^2 \ln(\varepsilon)]}{(a\varepsilon_s^2 \cdot \varepsilon^{(\varepsilon/\varepsilon_s)^2} - a\varepsilon^2 \varepsilon^{(\varepsilon/\varepsilon_s)^2} + b\varepsilon_s^2 \varepsilon)^2} + w_i \cdot \frac{(a - l\varepsilon^2)}{(a + m\varepsilon + l\varepsilon^2)^2} \quad (\text{A.20})$$

33 **(1.4.4) Considering the limit of $(1-w_i) \cdot P_l'(\varepsilon) + w_i \cdot S_o'(\varepsilon)$ as $\varepsilon \rightarrow 0^+$**

34 Combining Equations (A.10) and (A.16), we get

$$\lim_{\varepsilon \rightarrow 0^+} [(1 - w_i) \cdot P_l'(\varepsilon) + w_i \cdot S_o'(\varepsilon)] = (1 - w_i) \cdot E_0 + w_i \cdot E_0$$

$$= E_0 \quad (\text{A.21})$$

35 **(1.4.5) Extending the limit of $(1-w_i) \cdot P_l'(\varepsilon) + w_i \cdot S_o'(\varepsilon)$ as $\varepsilon \rightarrow \infty$**

36 Combining Equations (A.11) and (A.17), we get

$$\lim_{\varepsilon \rightarrow \infty} [(1 - w_i) \cdot P_l'(\varepsilon) + w_i \cdot S_o'(\varepsilon)] = 0 + 0$$

$$= 0 \quad (\text{A.22})$$

37 Combining sections from 1.1 to 1.4, we get Table A.1. This table summarizes the performance of

38 three models—Duncan-Chang's model, Jiang's model, and DC-JL model—across various reference

39 values. It shows that all models have the same E_0 value as ε approaches 0^+ , whereas only Jiang's and DC-

40 JL models have TSS values as ε approaches ε_s . However, Duncan-Chang's model is more excellent than

41 Jiang's model when ε is near the test end (Table A.1).

42 **Table A.1** Summary of theoretical analysis results

Strain (ε)	Ideal value	Duncan-Chang model [1]	Jiang's model [2]	DC-JL model (This study)
$\varepsilon \rightarrow 0^+$	Initial slope	E_0	E_0	E_0
$\varepsilon \rightarrow \varepsilon_f$	Strength	TSS	-	TSS
$\varepsilon \rightarrow \text{test end}$	Residual strength	RSS	TSS	$<< RSS$
$\varepsilon \rightarrow +\infty$	Ultimate residual strength	URS	TSS	0
Comparison		Worse	Limited	Advanced

2 A case of an application

This study demonstrates the application of the DC-JL model to a practical case (Table A.2). The parameter E_0 governs the slope of the stress-strain curve in the initial stage, which is determined by the slope value in the linear-elastic phase. TSS represents the maximum value of the stress-strain curve, corresponding to the peak strength. The yield strain (ε_s) is the point at which the soil transitions from linear-elastic deformation to plastic deformation. The parameter w_i is the comprehensive weight (%), defined as the ratio of the area between Area ① and the total area (the sum of Areas ② and ①). Area ① represents the area between the stress-strain curve and the plasticity term curve $P_i(\varepsilon)$, while Area ② represents the area between the stress-strain curve and the softening term curve $S_o(\varepsilon)$ (Figure A.1).

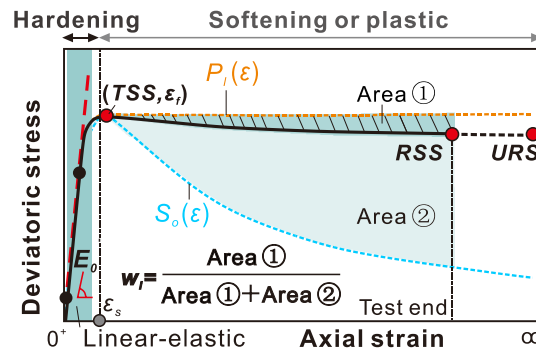


Fig. A.1 Key model parameters.

Table A.2 A case of experimental stress-strain curve data (<https://github.com/LuTuo123/DC-JL-model>)

ε (%)	$(\sigma_1 - \sigma_3)$ (kPa)	Model value (kPa)	ε (%)	$(\sigma_1 - \sigma_3)$ (kPa)	Model value (kPa)	ε (%)	$(\sigma_1 - \sigma_3)$ (kPa)	Model value (kPa)
0.001	0.00	0.22	5.500	258.18	256.97	11.000	246.11	244.17
0.500	133.18	105.80	6.000	256.41	255.01	11.500	245.52	243.58
1.000	224.54	212.75	6.500	254.83	253.29	12.000	244.97	243.03
1.500	259.93	265.37	7.000	253.42	251.77	12.500	244.46	242.52
2.000	275.03	275.03	7.500	252.16	250.42	13.000	243.99	242.06
2.500	273.19	273.71	8.000	251.02	249.22	13.500	243.55	241.62
3.000	269.60	270.91	8.500	250.00	248.15	14.000	243.14	241.22
3.500	267.25	267.71	9.000	249.07	247.19	14.500	242.76	240.84
4.000	264.75	264.59	9.500	248.23	246.32	15.000	242.40	240.49
4.500	262.35	261.74	10.000	247.46	245.54			
5.000	260.15	259.21	10.500	246.76	244.82			

As shown in Figure A.2 and Table A.3, the DC-JL model's calculated values closely match the experimental data. Additionally, the changes in the first derivative (slope of the experimental data) are clearly illustrated. Notably, only the derivative of the DC-TL model aligns with the correct slope.

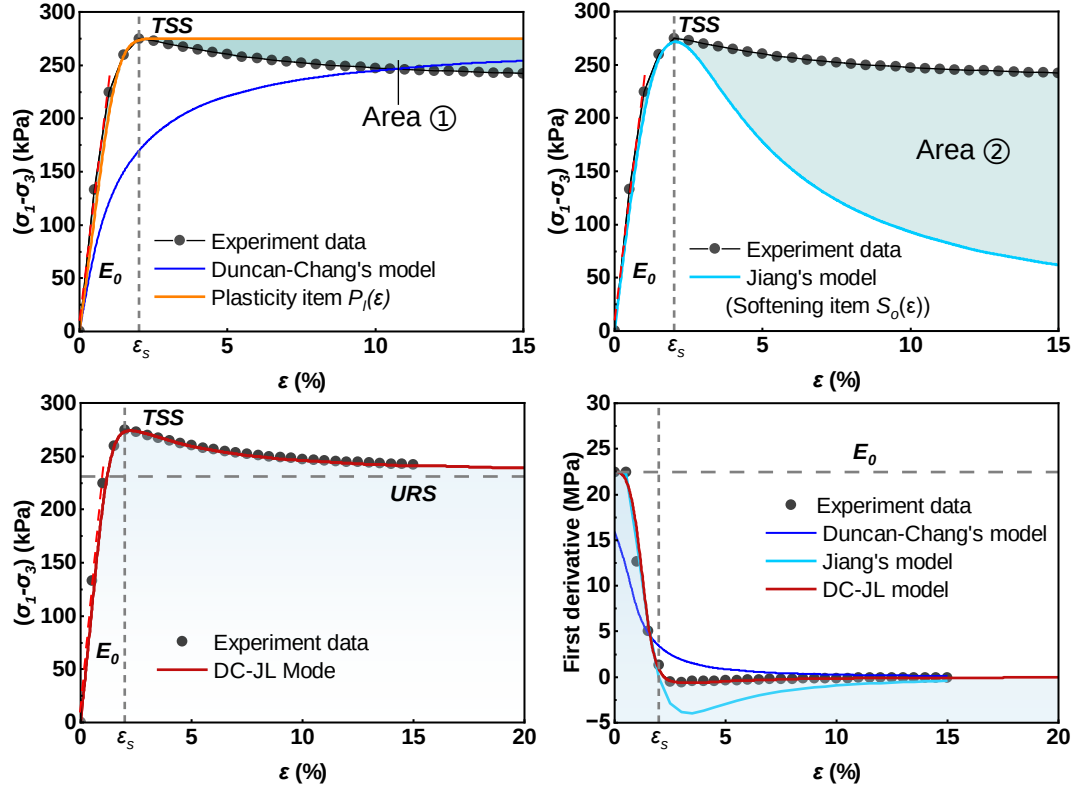


Fig. A.2 Experimental data vs. DC-JL model.

Table A.3 DC-JL model's parameters

E_0 (kPa)	TSS (kPa)	ϵ_f (%)	w_i (%)	ϵ_s (%)	R^2	URS (kPa)
22480	275.03	2.00	16.20	2.00	0.99	230.67

Reference

1. J. M. Duncan, and C.-Y. Chang. "Nonlinear analysis of stress and strain in soils," *Journal of the soil mechanics and foundations division* 96 (1970):(5) 1629-53.
2. Y. Jiang, X. Xian, and J. Su. "Research on distortion of single rock and constitutive relation," *Yantu Lixue(Rock Soil Mech)* 26 (2005):(6) 941-5, <https://doi.org/10.16285/j.rsm.2005.06.025>.